



MindSpark

– Where Fun Meets Focus



- **Team ID:** TC26-GF-065
- **Domain(s):** Electronics and Embedded Systems
- **SDG Goals:** SDG 4 – Quality Education
SDG 9 – Innovation & Technology
- **Scientific concept:** Interactive learning using color recognition and basic electronics

Team Lead: Vishwarupa Kokate

Scientific Concept: Memory & Pattern Recognition



Scientific concept

- Memory & Pattern Recognition

Domain

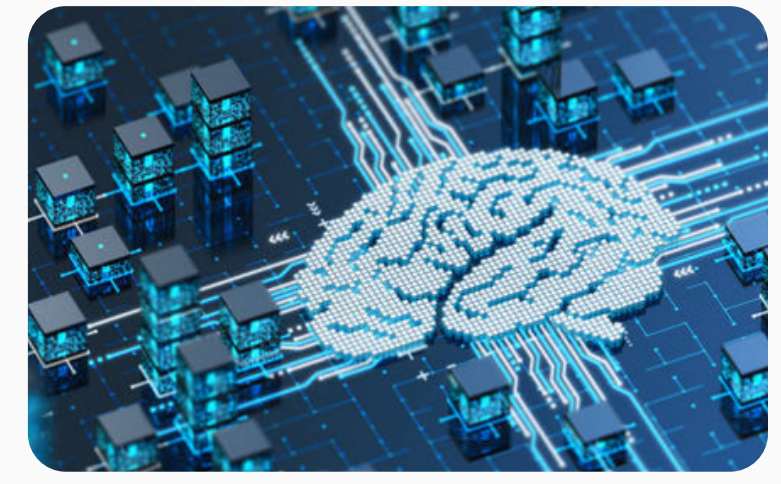
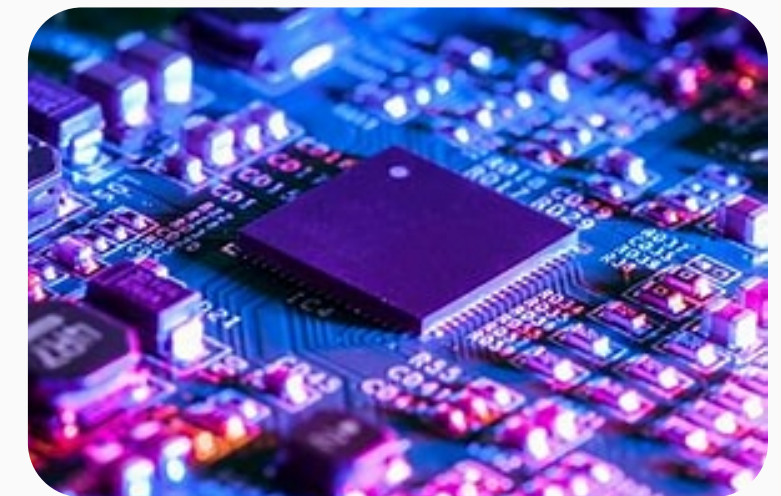
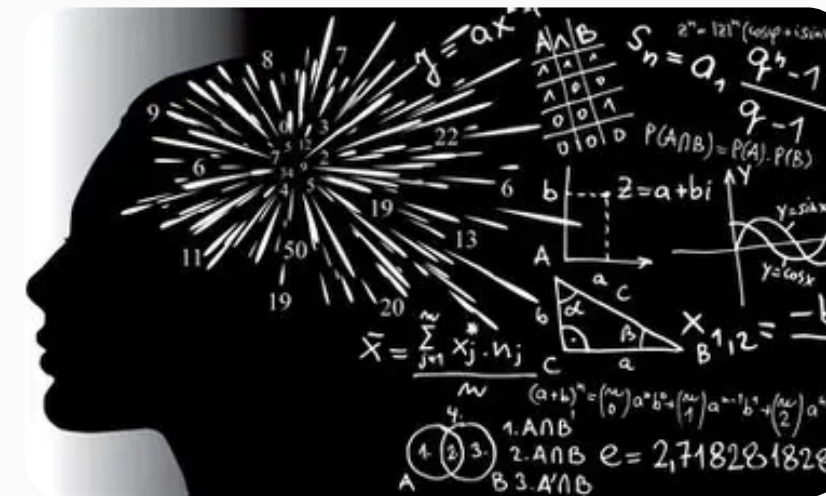
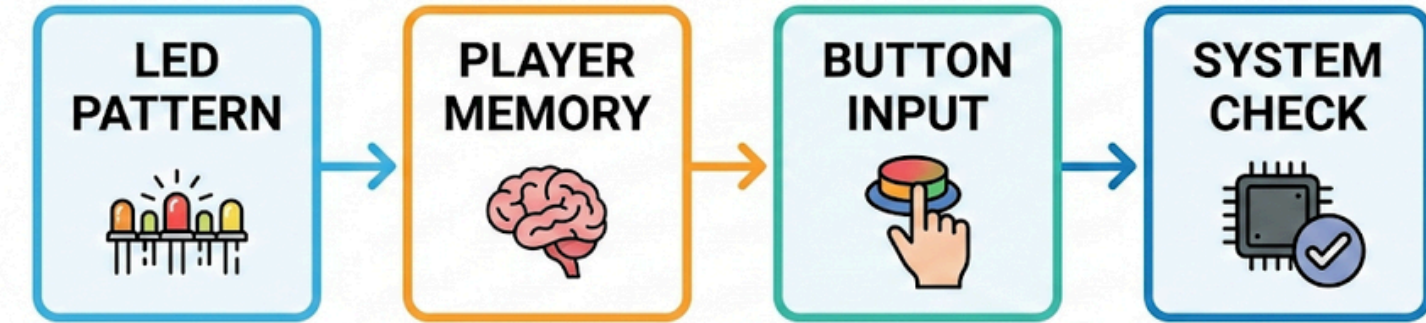
- **Electronics:** Interactive LED system
- **Mathematics:** Sequence and pattern logic
- **Cognitive Science:** Memory and pattern processing

Concept Overview

- LEDs display a **pattern sequence**
- Player observes and **memorizes** the pattern
- Player repeats the pattern using buttons
- System **verifies** the response

Real-Life Applications

- **Education:** Learning sequences like alphabets, numbers, and formulas
- **Security Systems:** Face recognition and fingerprint identification
- **Smart Technology:** Voice assistants recognizing speech patterns





Concept-to-Toy Mapping

Concept → Toy Mechanism

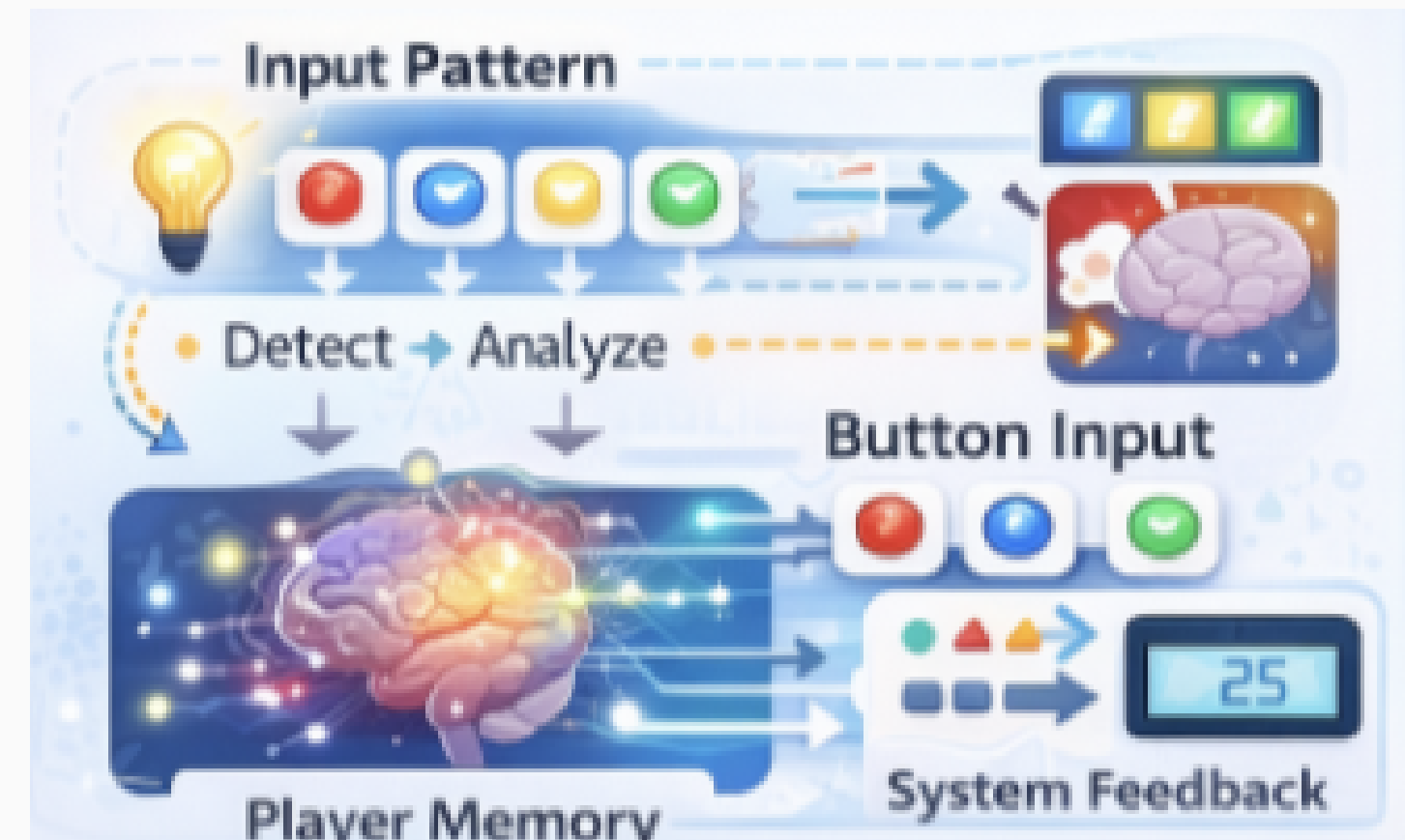
- LEDs display pattern sequence
- Player observes & repeats
- Arduino checks input & increases level

Learning Objectives

- Memory retention
- Pattern recognition
- Reaction speed
- Basic electronics

STEM Concepts Learned

- How memory and pattern recognition work
- Input → Processing → Output in electronic systems
- Learning through interactive STEM gameplay



Toy Overview



MindSpark

AI Adaptive Memory & STEM Learning Toy

Category:

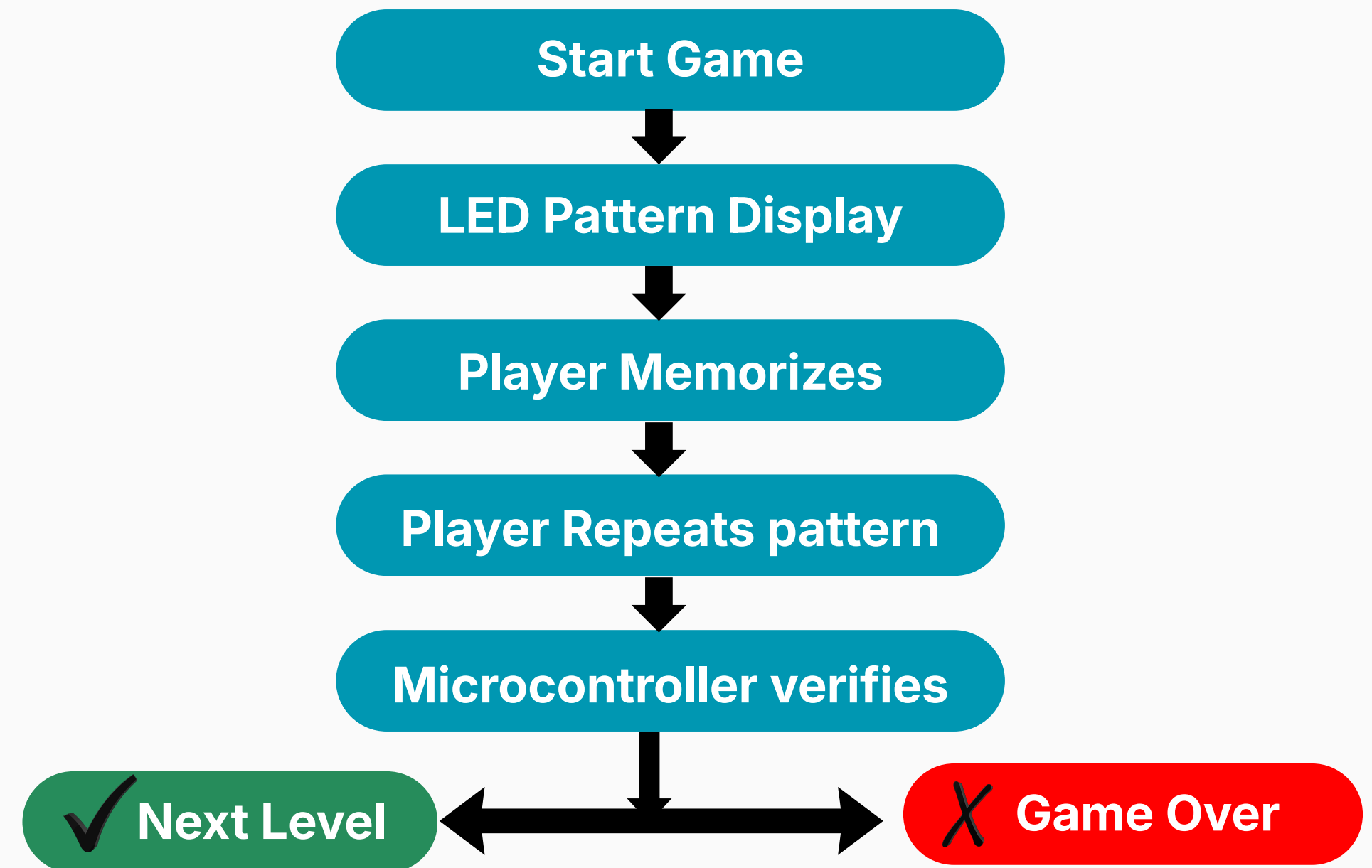
- STEM Educational Toy
- Electronics & Embedded Systems Toy

Game modes:

- Multiple **interactive game modes** designed to improve memory, logic, and attention.
- Players analyze LED **patterns & puzzles**
- **Increases complexity** based on player performance.

User-Toy Interaction

- **Visual:** Observe LED pattern
- **Physical:** Press buttons
- **Feedback:** Buzzer + LCD score



Working Mechanism (Graphical Explanation)



INPUT: Player interacts with the game through buttons.

- Push Buttons – Player repeats the LED sequence by pressing corresponding buttons.
- Reset Button – Restarts the game and clears level/score.

PROCESSING

The Arduino Nano acts as the brain of the system.

Functions performed:

- Generates random LED patterns
- Stores sequence in memory
- Reads player button inputs
- Compares player input with stored pattern
- Controls level progression and difficulty

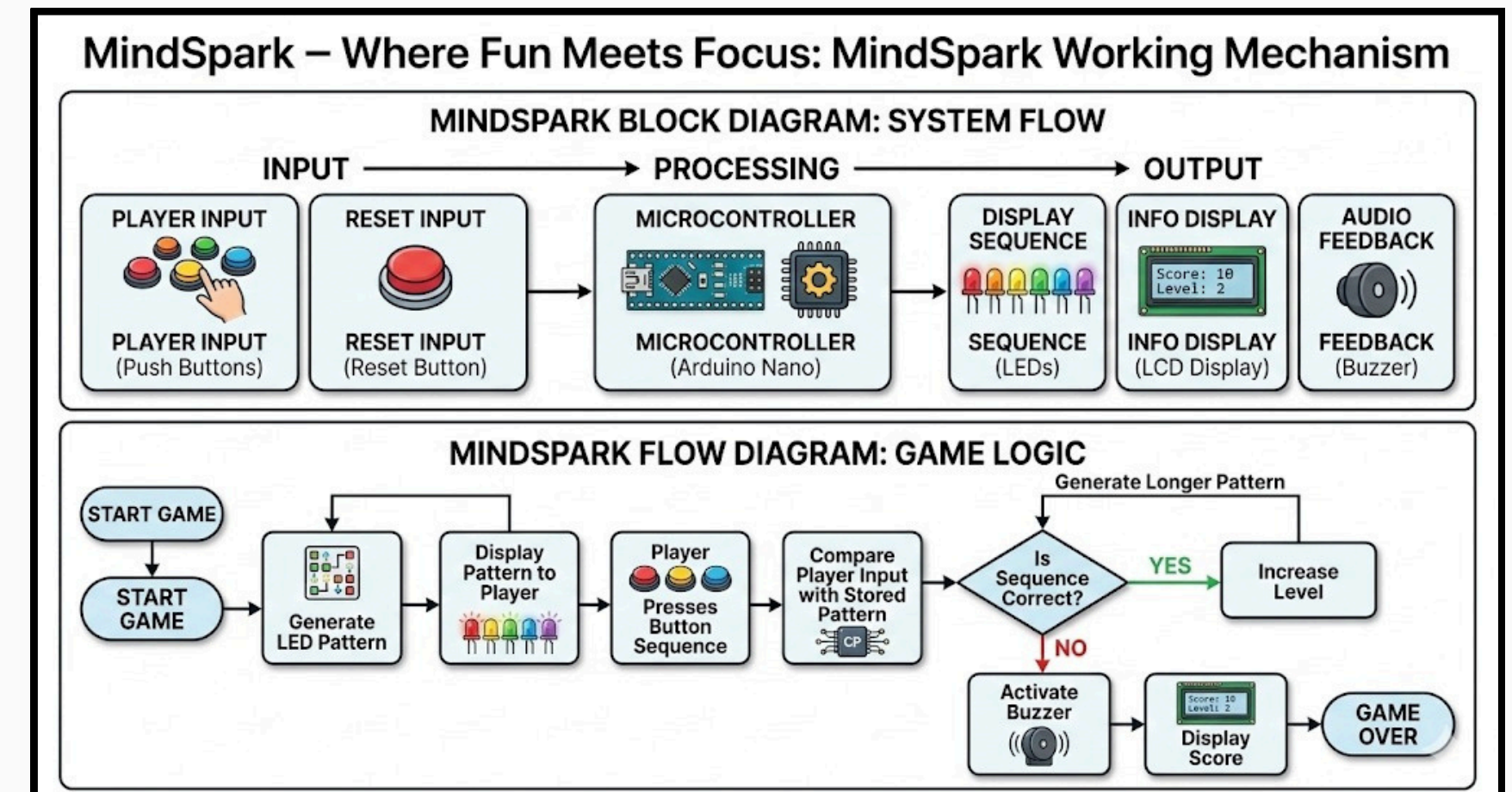
OUTPUT

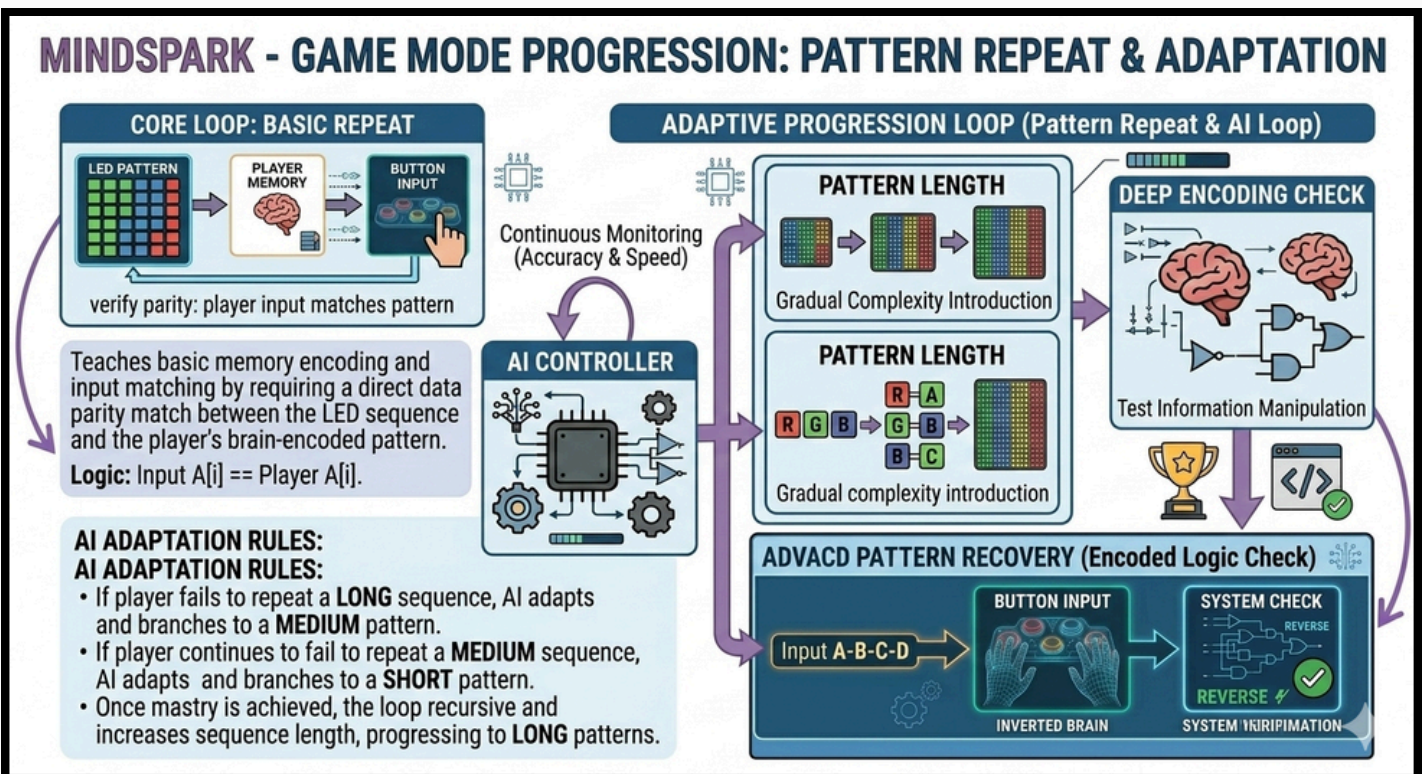
The system provides feedback to the player.

- LEDs – Display the pattern sequence to memorize

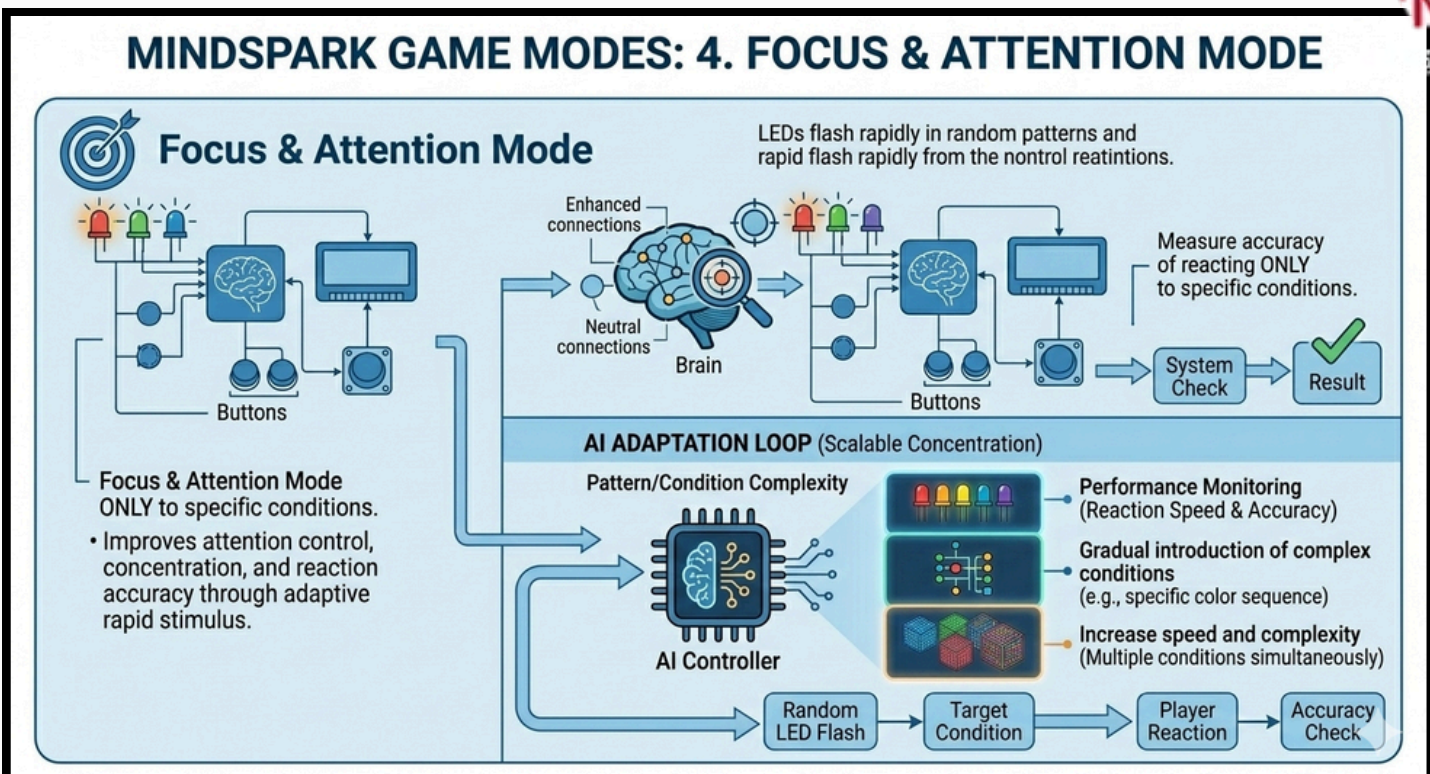
Example: → → →

- LCD Display – Shows score, level, and game status

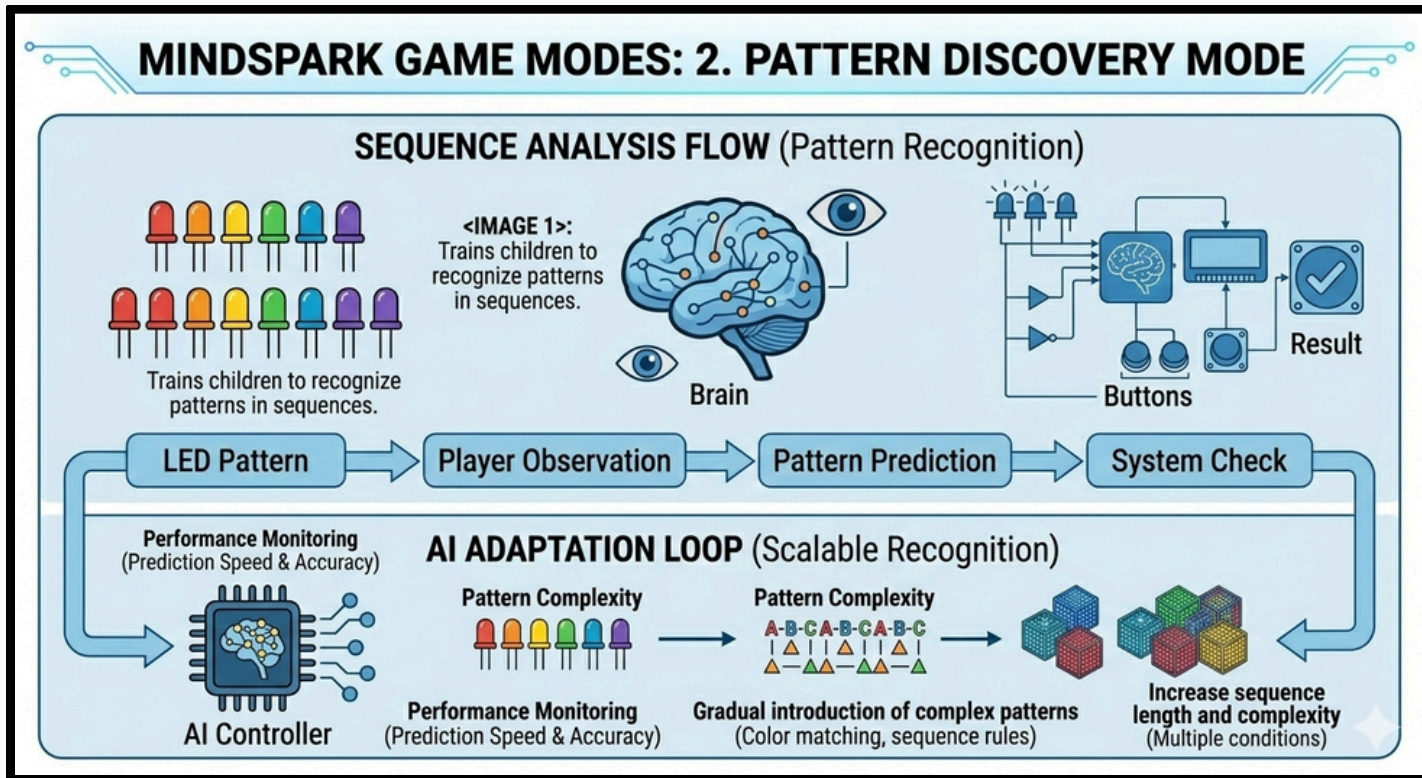




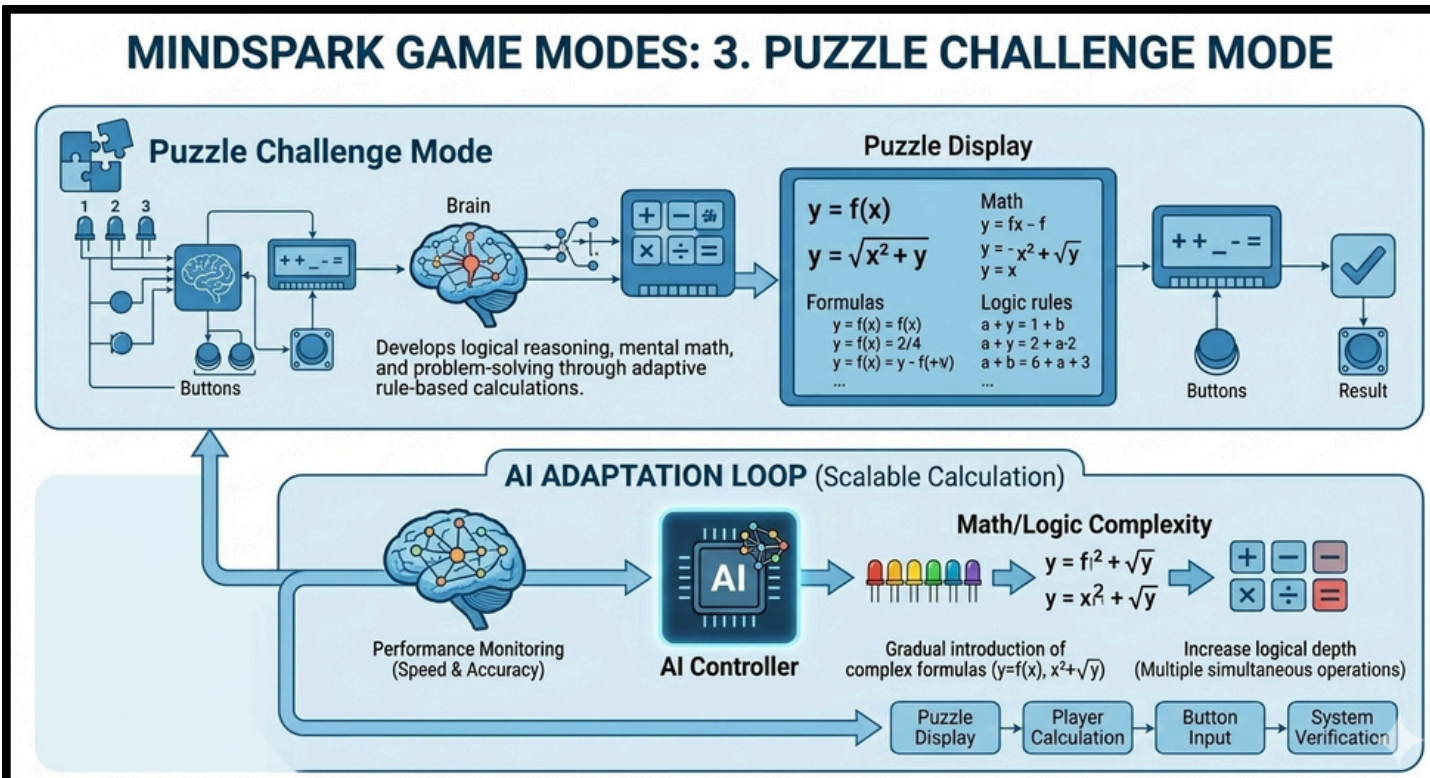
MODE I: Core Pattern Loop (Adaptive Repeat)



MODE IV: Focus & Attention Mode (Reaction Control)



MODE II: Pattern Discovery Mode (Sequence Analysis)



MODE III: Puzzle Challenge Mode (Logic & Calculation)

MindSpark Hardware Architecture



1. Core Processing Unit

- Arduino Nano / ESP32 acts as the central controller.
- Processes inputs, generates patterns, and runs game logic.

2. Input Interface

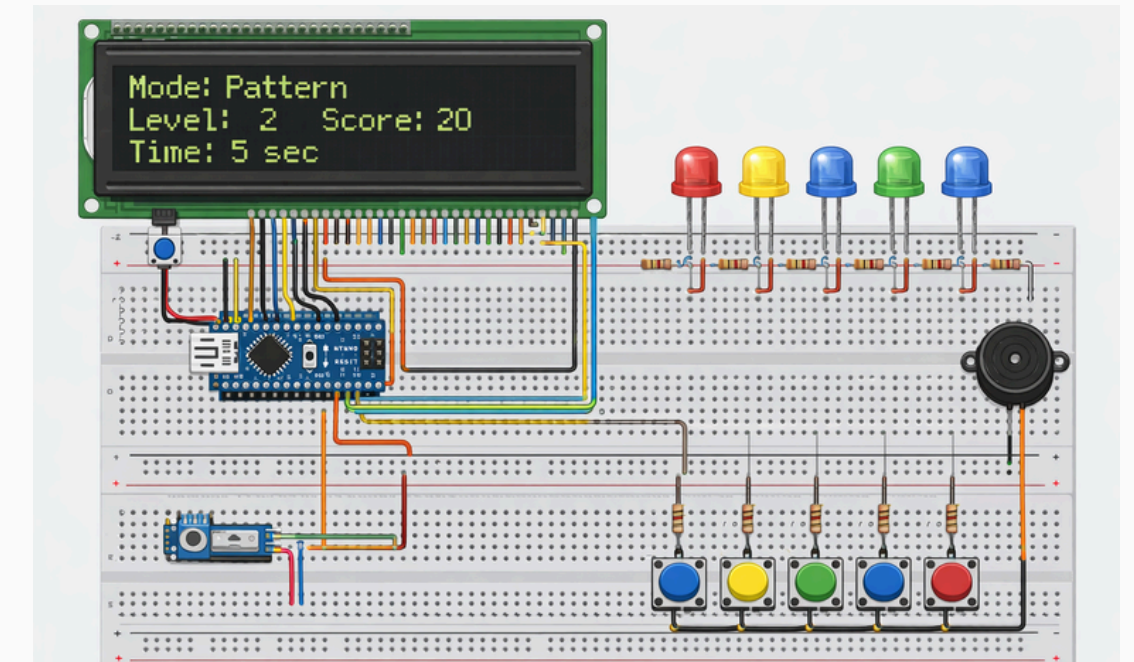
- Push buttons, potentiometer, and reset button capture player actions.

3. Output Interface

- Colored LEDs and buzzer provide visual and audio feedback
- 20x4 LCD displays game information and progress.

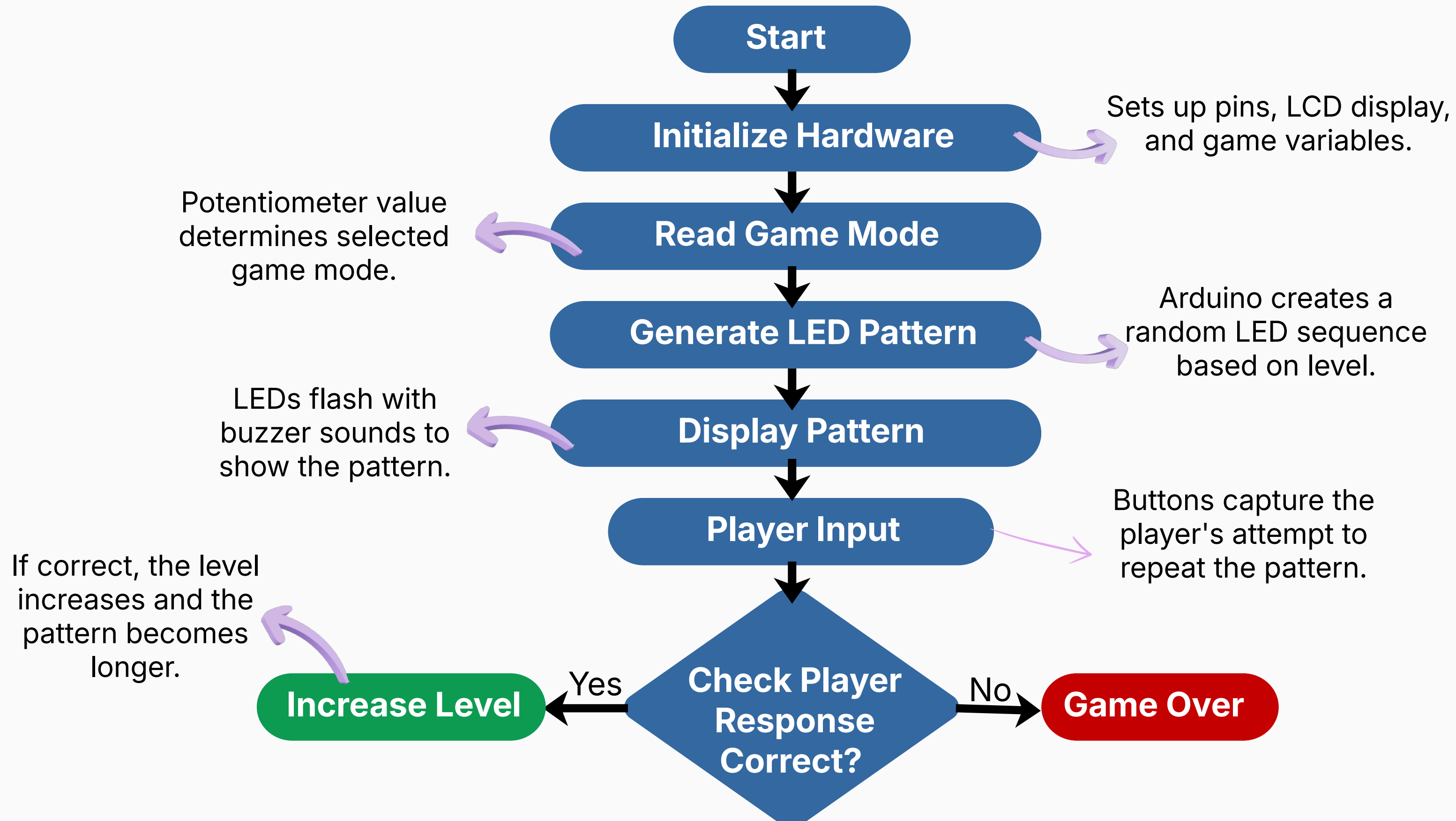
4. Power & Integration

- Powered by 5V USB through Arduino Nano / ESP32.
- Dotboard and resistors ensure safe circuit connections.



"The hardware architecture is designed to provide an interactive learning experience by combining visual, audio, and tactile feedback."

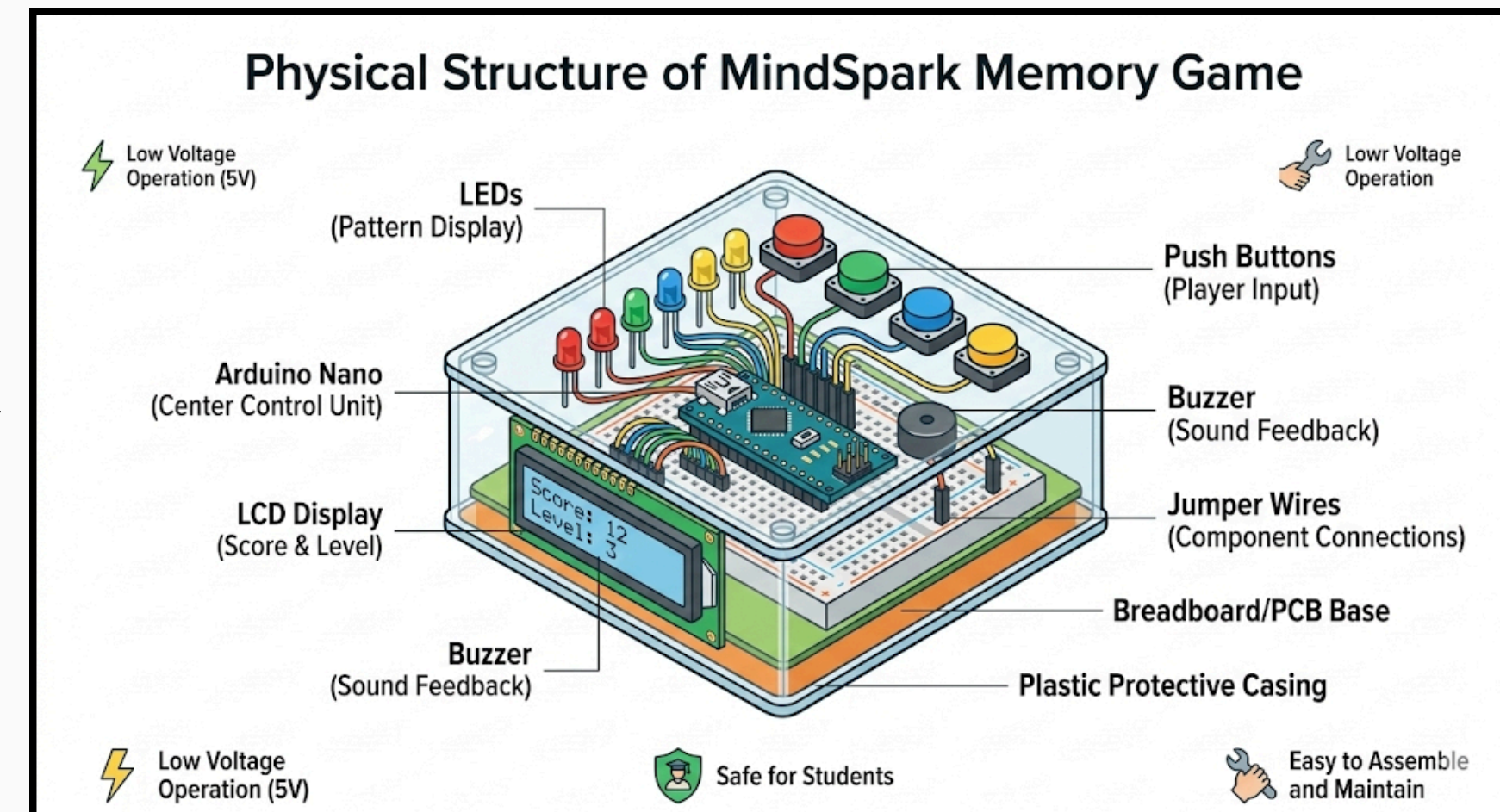
MindSpark Software Architecture



Hardware & Detailed Toy Design

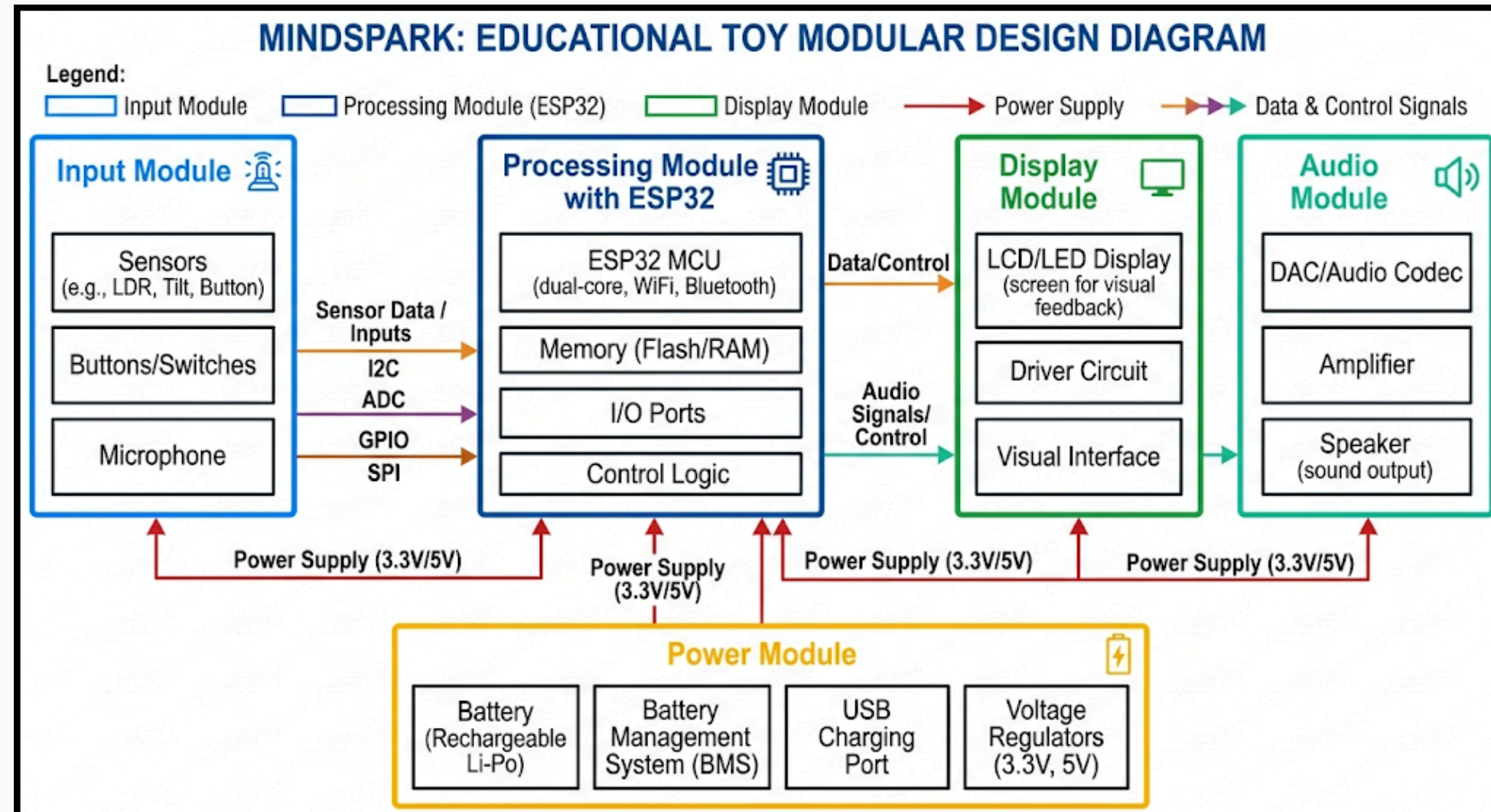


- **Physical structure:** consisting of an Arduino board, LEDs, push buttons, buzzer, and LCD display mounted on a small board or box.
- **Built using** safe and lightweight materials such as plastic casing, jumper wires, breadboard/PCB, and electronic components.
- **Safety considerations:** low voltage operation (5V), properly insulated wiring, and smooth edges to prevent injury during use.
- **Designed** for easy assembly and maintenance, allowing students to connect components and understand the circuit easily.
- **Simple to use:** players only observe LED patterns and press the corresponding buttons to repeat the sequence.



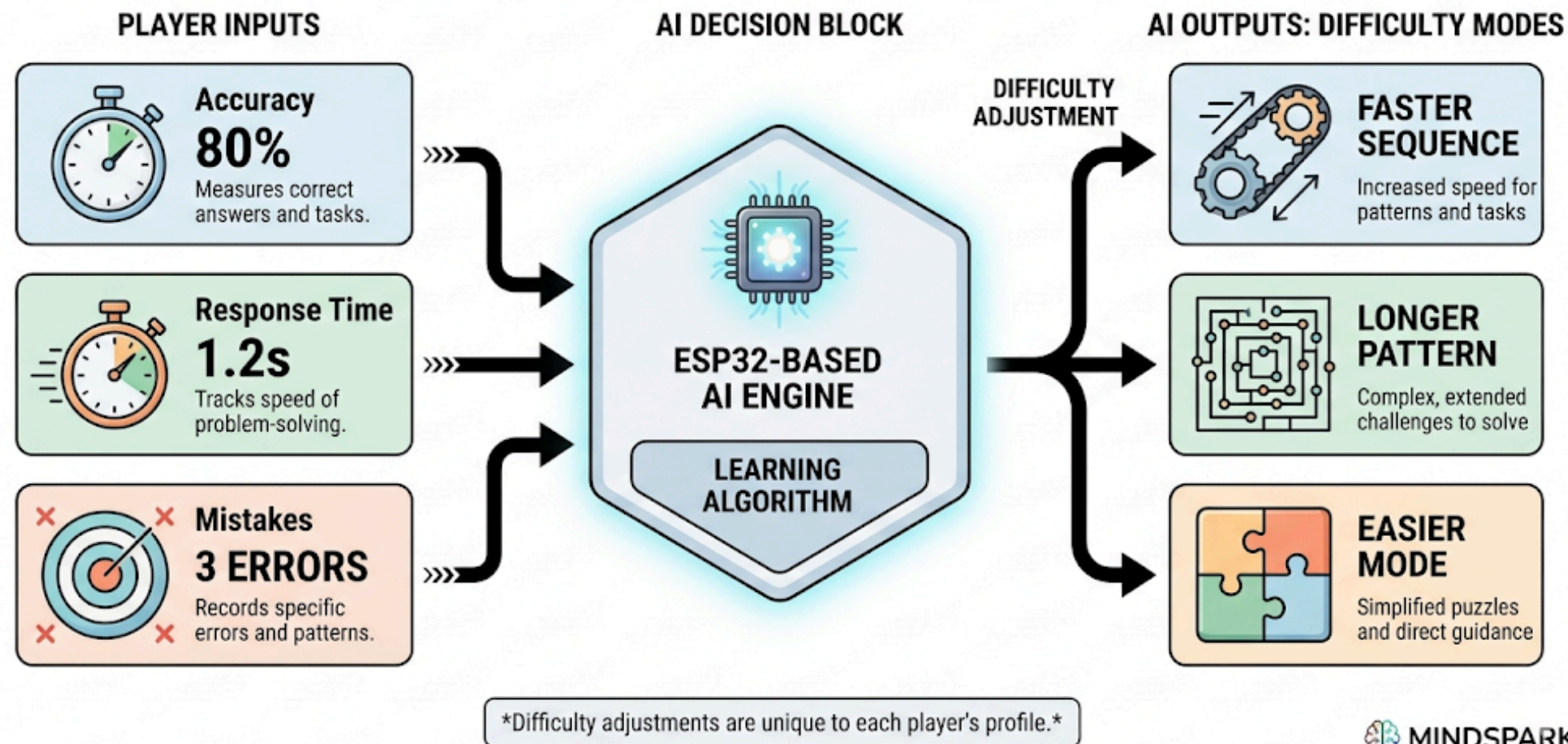
Conceptual representation of the hardware structure of the MindSpark Memory Game built using an Arduino Nano and interactive electronic components.

Modular Design



- **ESP32** acts as the central processing unit controlling all game operations.
- **Input module** captures user actions through buttons and optional sensors.
- **Display module** provides visual feedback using LEDs and LCD interface.
- **Audio module** generates buzzer/speaker feedback for game interaction.
- **Power module** ensures stable 3.3V / 5V supply for safe operation.

MINDSPARK AI ADAPTIVE LEARNING: TAILORING CHALLENGES

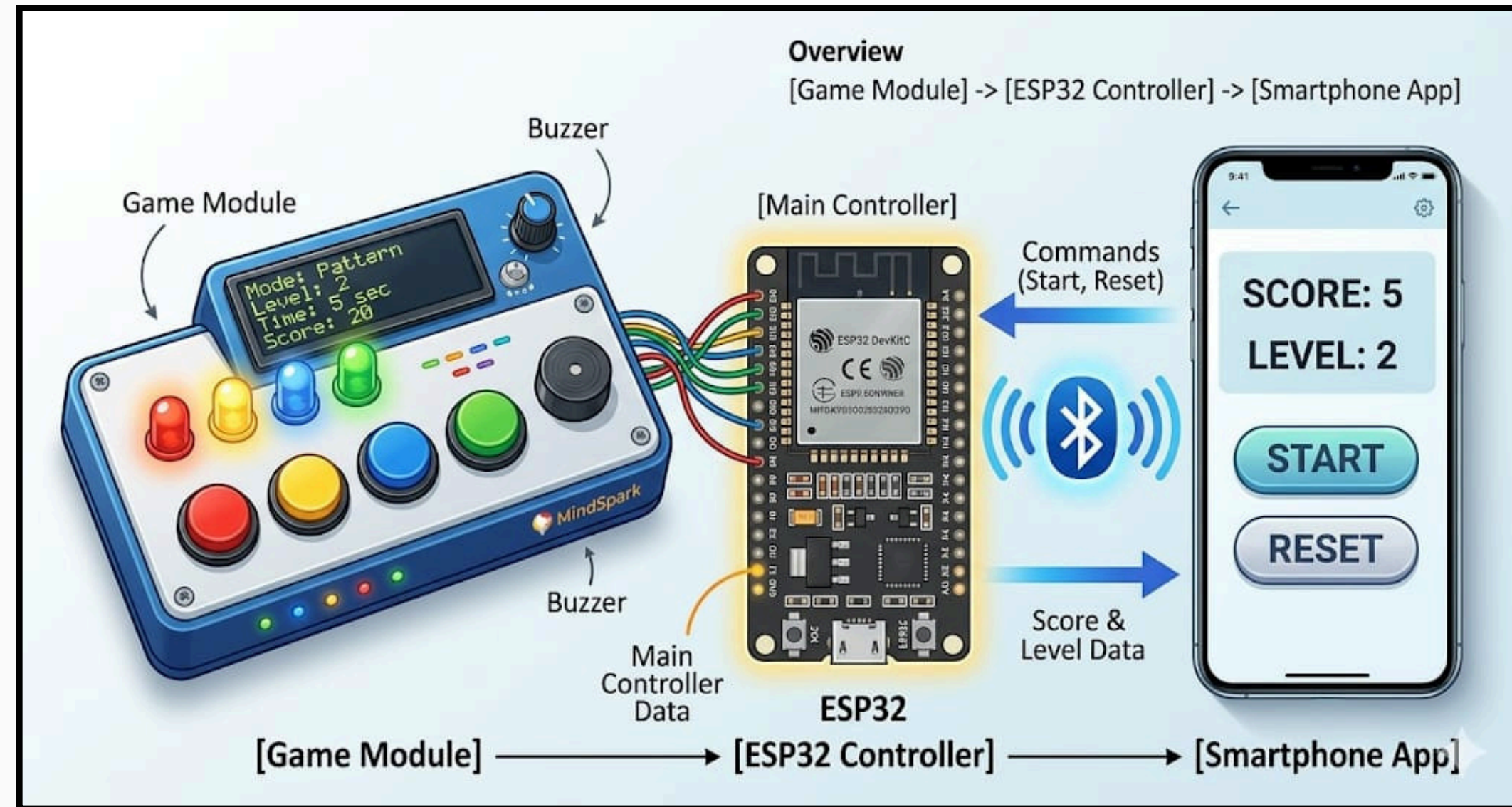


- **Mode 1** – Core Pattern Loop, if the player performs well, the LED sequence becomes longer and the blinking speed increases.
- **Mode 2** – Pattern Discovery Mode, AI introduces more complex sequence arrangements or missing-pattern challenges when correct responses are consistent.
- **Mode 3** – Puzzle Challenge Mode, higher performance leads to more logical combinations and reduced solving time, while repeated mistakes simplify the puzzle.
- **Mode 4** – Focus and Attention Mode, AI changes reaction speed by reducing the response window for faster players and increasing it when errors are frequent.

Mobile Integration Features



- Bluetooth communication using ESP32
- Real-time score and level tracking
- Remote control commands
- Performance data logging
- Future adaptive learning support



Target Age Group & Educational values



Target Age Group: (8 - 16)

- Stage where **logical reasoning** and **analytical thinking** develop.
- Students can understand **patterns, sequences**, and **cause-effect** relationships.
- Encourages early interest in **STEM** and technology.



Educational Value

- Improves memory, **focus**, and **pattern recognition** through **LED-based** challenges.
- Develops logical thinking and problem-solving skills.
- Introduces basic STEM concepts like **electronics and programming logic**.
- **Adaptive difficulty levels** help students progressively improve cognitive skills.



Innovation & Uniqueness



AI Adaptive Learning

Difficulty adjusts automatically based on player performance.

STEM Learning

Introduces Arduino programming, electronics, and system logic.

Cognitive Skills

Improves memory, focus, and pattern recognition.

Expandable Hardware

Supports Bluetooth, Wi-Fi, RGB LEDs, and sensor modules.

AI Adaptive Learning

Teachers and students can modify code, speed, and difficulty levels.

"An interactive toy that combines AI adaptation, cognitive training, and STEM learning in a single platform."



References & Resources

References:

1. Product Drawings / Sketch References
 - [Arduino Simon Says Game Tutorial \(Instructables\)](#)
1. Circuit & Hardware Learning Resources
 - [Arduino Memory Game Project \(Arduino Project Hub\)](#)

Resources:

1. GitHub Repository
 - <https://github.com/Vishwarupa3/MindSpark-STEM-Toy>
2. TinkerCad Simulation
 - [TinkerCad](#)

Thank You!



Team Members

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